

Unit 10: Sedimentary Rocks

Reading: pages 134–150 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. *Lithification* is
 - a) the process by which compacted sediments become cemented together into solid rock
 - b) the chemical weathering of an exposed rock face
 - c) the melting of sediment to form magma
 - d) the movement of sediment from the site of erosion to the site of deposition
2. Most sand-sized clasts are made of quartz because
 - a) quartz dissolves readily and reprecipitates as sand
 - b) sand can only form from igneous rock
 - c) quartz is the only mineral that breaks into sand-sized pieces
 - d) quartz is more resistant to weathering than any other common mineral
3. A clastic sedimentary rock dominated by *rounded* clasts larger than 2 mm is
 - a) conglomerate
 - b) arkose
 - c) shale
 - d) breccia
4. By far the most common chemical sedimentary rock is
 - a) Chert
 - b) Limestone
 - c) Halite
 - d) Banded iron formation
5. As an inland sea in an arid region evaporates, which of these precipitates *last*?
 - a) Carbonates, at about 50% of the original volume
 - b) Gypsum, at about 20% of the original volume
 - c) Halite, at about 10% of the original volume
 - d) They all precipitate at the same time

6. Cross-bedding is especially valuable to geologists because it
- a) contains more fossils than any other structure
 - b) records the direction of current or wind flow at the time of deposition
 - c) proves a rock is chemical rather than clastic
 - d) gives the exact numerical age of the rock

Part B · True or False

2 questions × 1 point = 2 points

7. The principle of inclusions states that rock fragments within a sedimentary layer must be older than the layer itself.

TRUE / FALSE

8. Most fossils are found in igneous rock.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain the difference between *clastic* and *chemical* sedimentary rocks, and give one example of each.

10. Banded iron formations record a major change in Earth's atmosphere. Explain what that change was, and why these rocks essentially stopped forming after about 1800 Ma.

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